

CS 5393 FINAL

RECURRENT NEURAL NETWORKS AND STOCK PREDICTION

AGENDA

Project Overview

Models: LSTM vs GRU

Features

Activation Functions

Model Performance and Comparison

Applications and Key takeaways

PROJECT OVERVIEW

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- In this project, I used recurrent neural networks, or RNNs, to predict NVIDIA stock prices
- Each model was given data using **yfinance**, an open-source tool that uses Yahoo's API to retrieve stock information
- Model performance evaluated using **Root Mean Squared Error** and **Mean Absolute Error**

PROJECT OVERVIEW: EVALUATION METRICS

Metric	What it measures	Trading Relevance
Root Mean Squared Error (RMSE)	- Magnitude of large errors	Useful for risk management (i.e. preventing catastrophic predictions)
Mean Absolute Error (MAE)	- Average prediction error	Tells traders the "expected slippage", or error, of the model when acting on predictions

PROJECT OVERVIEW: TOOLS



LSTM VS GRU MODELS

LSTM VS GRU MODELS

- In this project, I trained both LSTM (Long Short-Term Memory) and GRU (Generated Recurrent Units) models
- Differences in architecture:
 - LSTMs have separate memory cell, input, output, and forget gates
 - GRUs combine input and forget gates into an update gate, simplifying its structure

LSTM VS GRU MODELS

- LSTM RNNs take longer to train and require longer computational times than GRU models
- LSTMs can be more effective at handling long-term dependencies than GRUs
- Both types of RNNs solve the vanishing gradient problem

FEATURES

FEATURES

Feature	Captures	RNN Benefit
Closing Price (Target)	Autoregressive Patterns	Used for time-series autoregression
SMA 50	Short-Term Trends	Smooths noise
SMA 200	Long-Term Trends	Smooths noise
Volume	Trade activity	Adds additional contextual information
Volume Change	Spikes in trade activity	Adds additional contextual information
14-Day Relative Strength Index	Overbought and Oversold stocks	Non-linear patterns for model to learn
Moving Average Divergence Convergence	Momentum	Non-linear patterns for model to learn

ACTIVATION FUNCTIONS

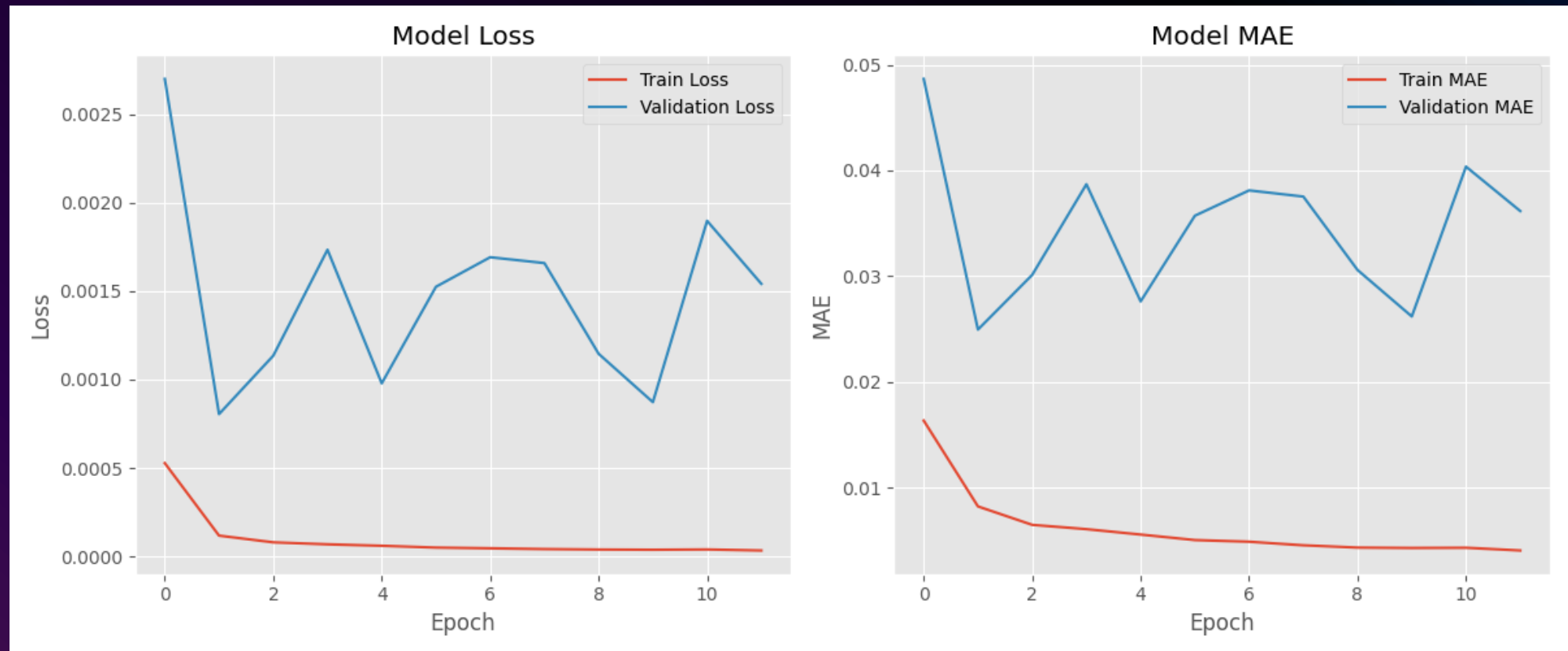
ACTIVATION FUNCTIONS: RELU

- Benefits of using ReLU for stock prediction:
 - Unlike Tanh or Sigmoid, ReLU does not saturate for positive values, mitigating the "vanishing gradient" problem
 - **Improved computational efficiency** over other methods
- Other activation functions:

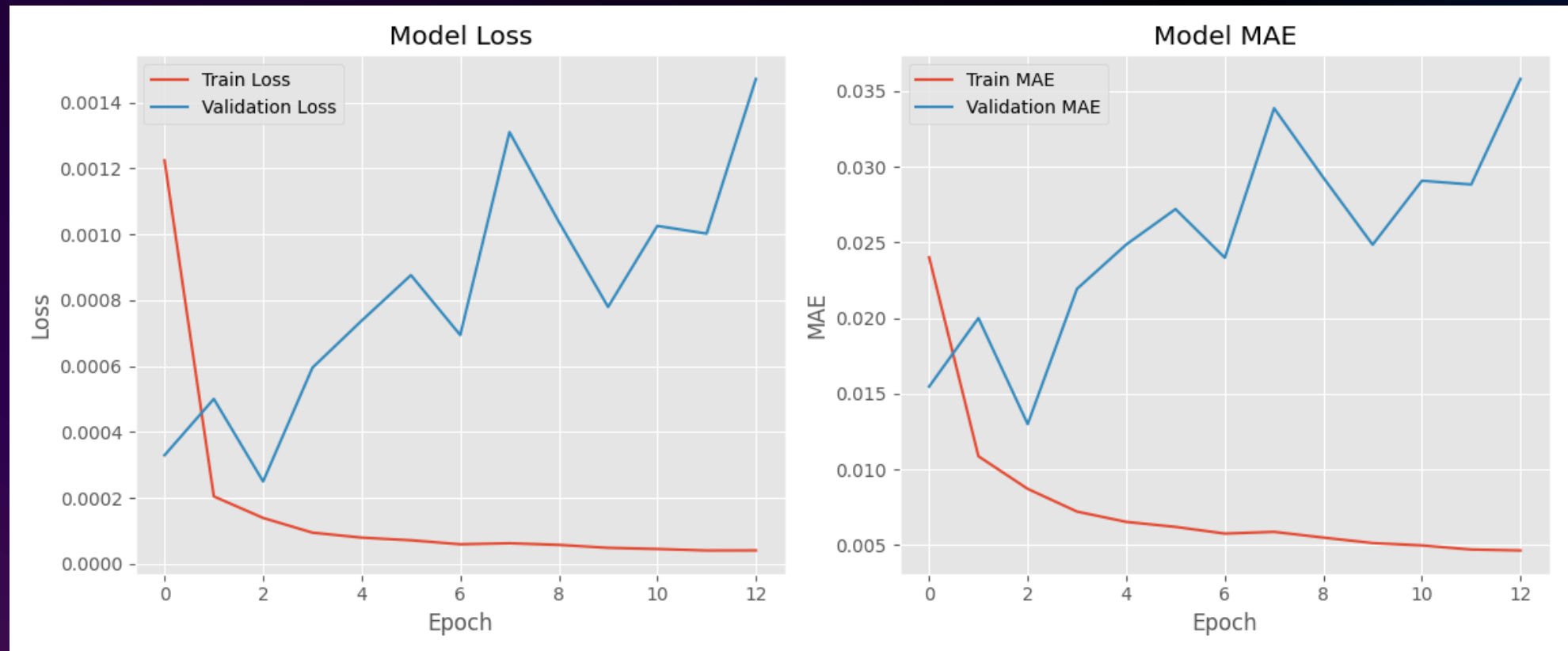
Activation Function	Potential Disadvantage
Sigmoid	Saturates for extreme values
Tanh	Computationally heavier than ReLU
Leaky ReLU	Computationally heavier than ReLU

MODEL PERFORMANCE

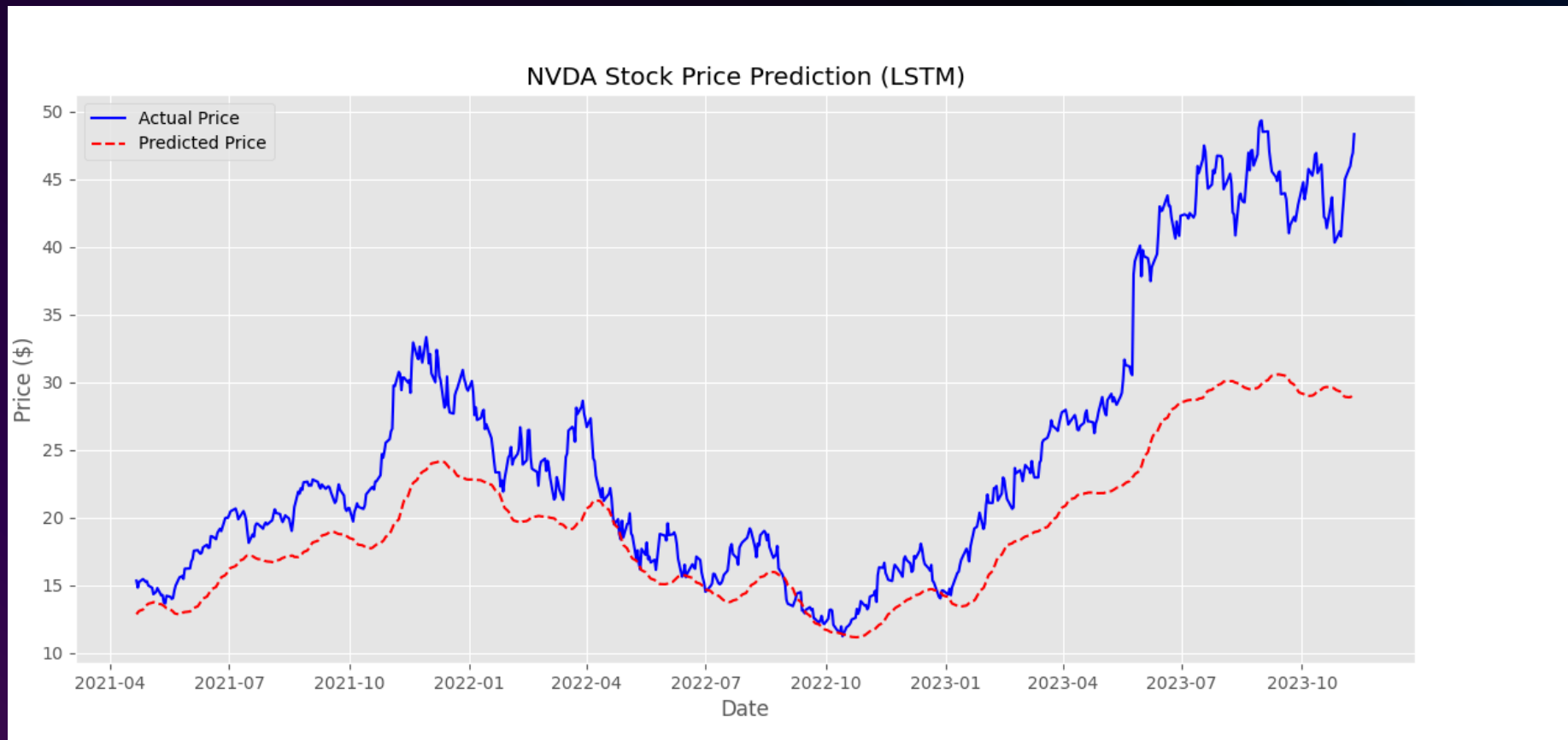
PERFORMANCE: LSTM LOSS



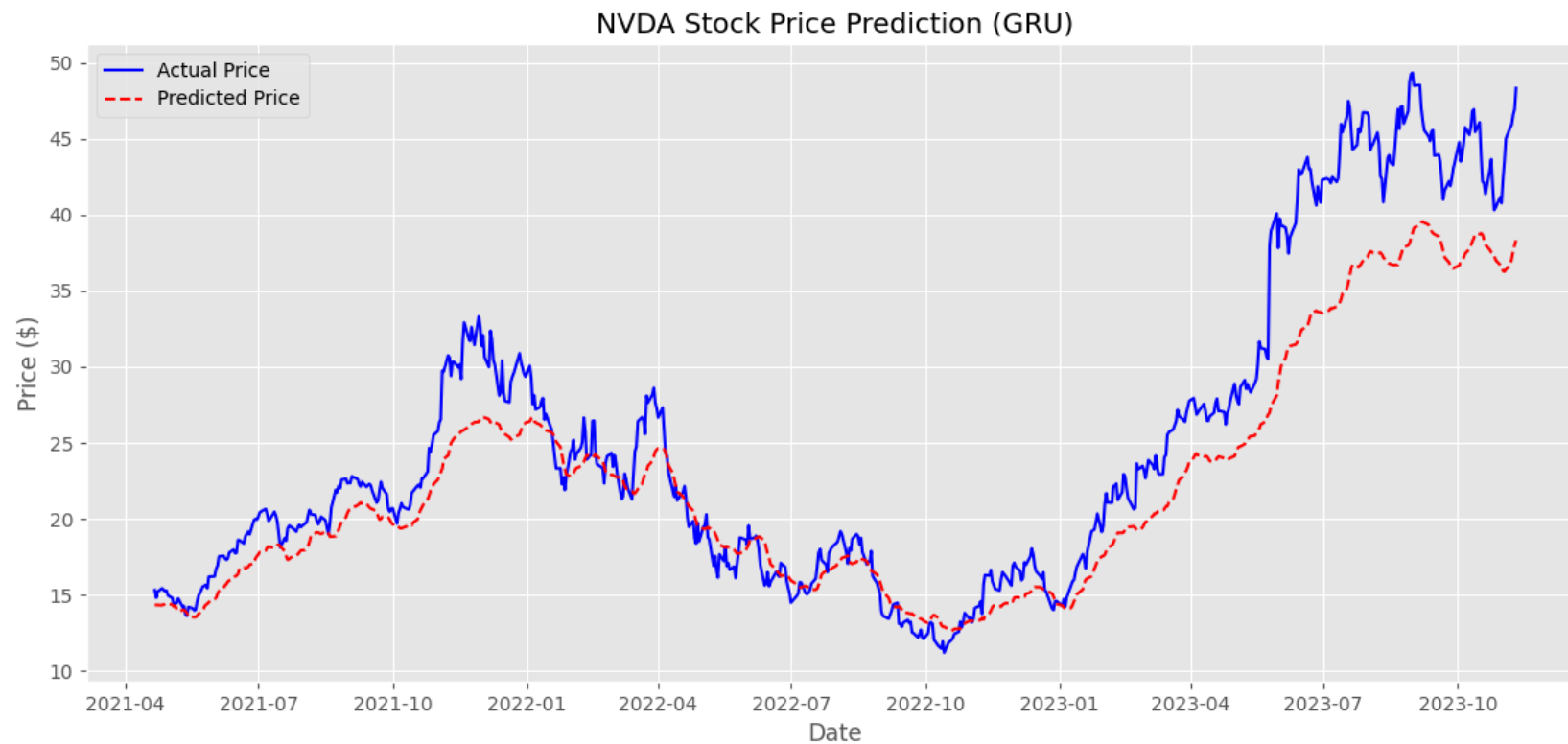
PERFORMANCE: GRU LOSS



PERFORMANCE: LSTM PREDICTION



PERFORMANCE: GRU PREDICTION



PERFORMANCE: LSTM VS GRU

PERFORMANCE METRICS COMPARISON

Metric	LSTM	GRU
RMSE	7.4966	4.0452
MAE	5.6389	2.9581
Training Time (s)	30.26	41.78
Testing Time (s)	0.5239	0.6693

KEY TAKEAWAYS AND APPLICATIONS

CONCLUSIONS: LSTM VS GRU

- GRU **outperformed** the LSTM model on NVIDIA stock prediction
- Why is this?
 - GRU can have comparable performance to LSTM on smaller datasets (Chung et al. (2014))
 - GRUs adapt to changes in data faster than LSTMs, and NVIDIA is a volatile stock

CONCLUSIONS: LSTM VS GRU

- GRU was **slower** than LSTM
- Why is this?
 - Early stopping: GRU was trained for more epochs than LSTM
 - GRUs adapt to changes in data faster than LSTMs, and NVIDIA is a volatile stock

CONCLUSIONS: APPLICATIONS OF MY MODEL

- Would I use my models in a real world-scenerio? **No**
- Better models exist
 - In 2024 a group of computer scientists created a LSTM that predicted HFDC stock with a MAE value of 0.21 (Barua et al. (2024))
- **Transformers** might create a more reliable model (long term)

CONCLUSIONS: NEXT STEPS

- Additional Testing, hyper parameter tuning
- Exploit NVIDIA's volatility with new features (Bollinger Band Width)

THANK YOU

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